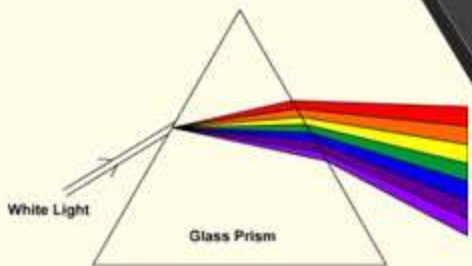


PHYSICS

FORCE – THE CONCEPTS

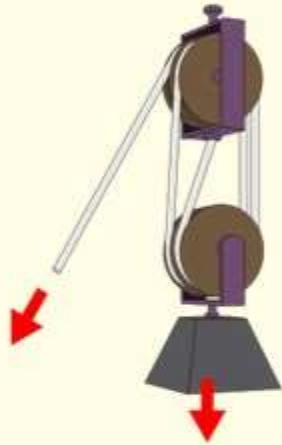
MODULE - 9



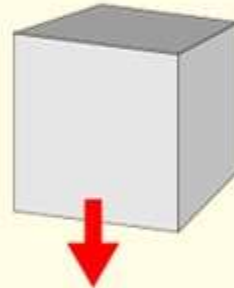
WHAT DO THESE PICTURES SUGGEST ?



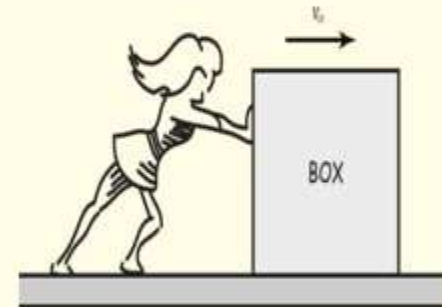
A pair of magnets



Pulley carrying load



A box lying on the ground



Pushing a heavy load

Some interaction is happening between two objects

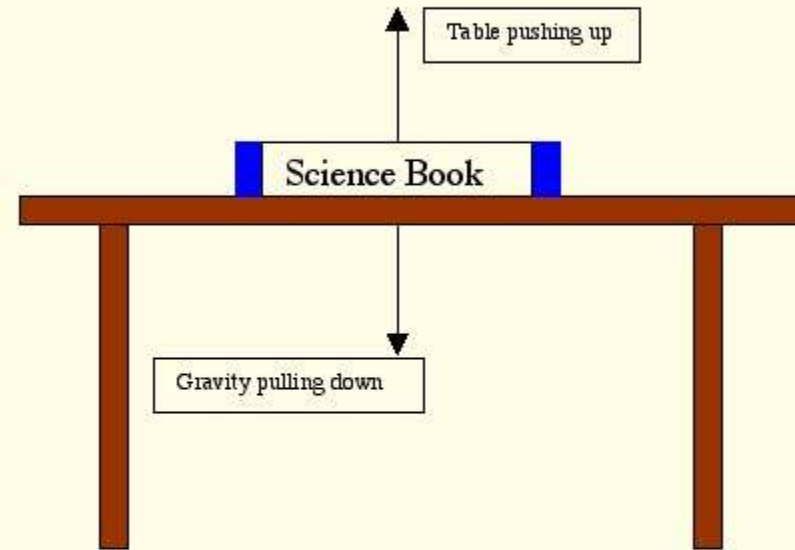
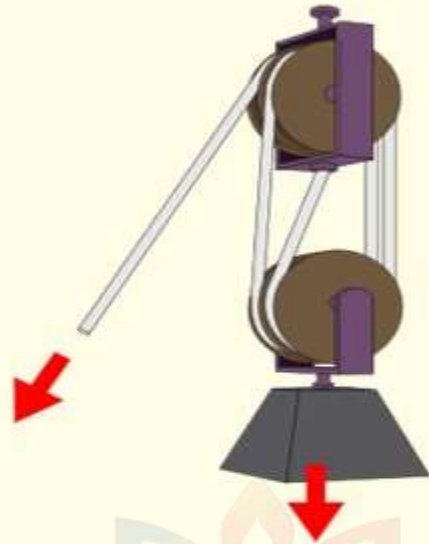
FORCE – AN INTERACTION



- Force is an interaction between two objects.
- A force can be a push or pull to any of the two objects or both.
- Force acts on each of the two objects during the interaction.
- As the interaction ceases, the two objects no longer experience the force.



FORCE ALWAYS HAS A DIRECTION



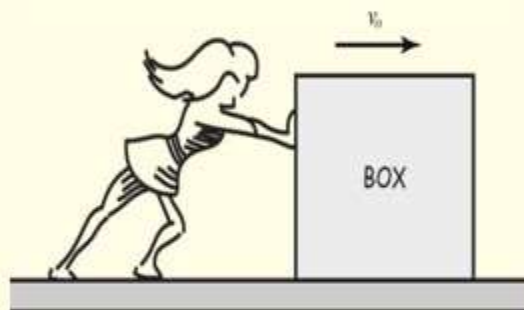
*Force always acts on an object in some direction. Thus,
Force is a Vector quantity.*



PHYSICS

FORCE

CONTACT FORCE



*The two interacting
bodies are in
contact*

NON-CONTACT FORCE



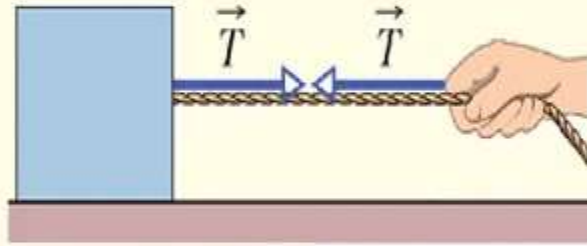
*The two interacting
bodies are not in
contact*

TYPES OF CONTACT FORCES

Frictional Force



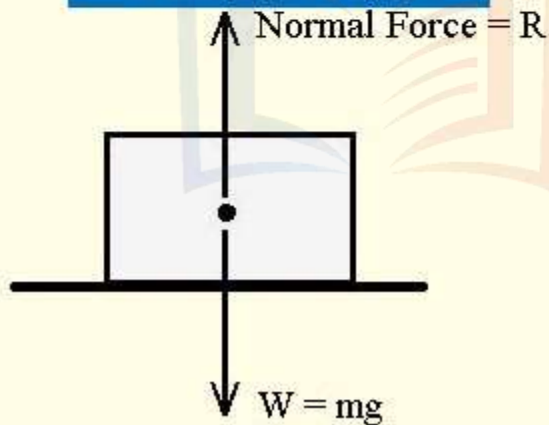
Tension Force



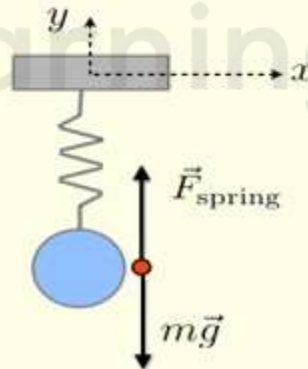
Applied Force



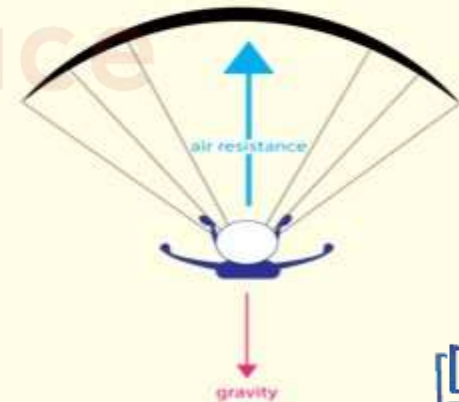
Normal Force



Spring Force

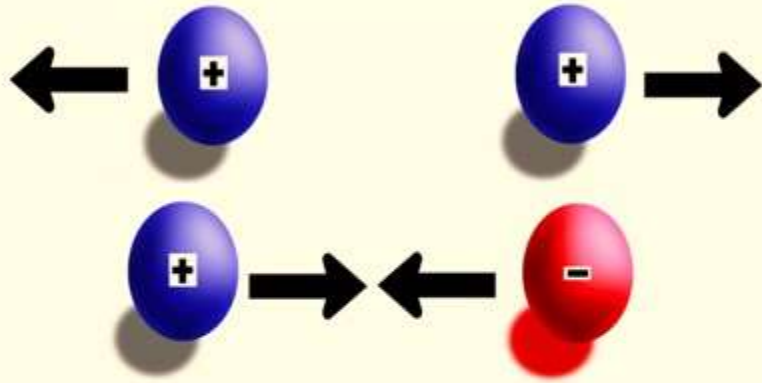


Air-resistance Force



TYPES OF NON - CONTACT FORCES

Electrostatic Force



Gravitational Force



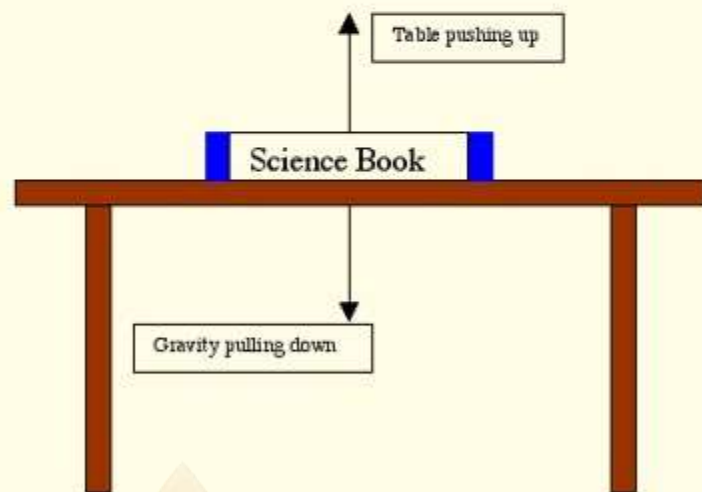
Magnetic Force



PHYSICS

FORCE

BALANCED FORCE

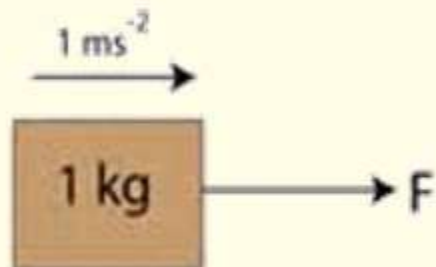


UNBALANCED FORCE



*In case of unbalanced forces, the body experiences movement.
Unbalanced forces can also bring acceleration like kicking a moving ball*

UNITS OF FORCE



- The force required to give an acceleration of 1 ms^{-2} to a mass of 1 kg is called 1 Newton (N) .
- The unit Newton finds applications in calculating energy, work and heat. It is also used to measure pressure, power and torque.
- Some other units of Forces, not commonly used are Dyne, poundal and pound.

A poundal is the force required to give a mass of 1 pound an acceleration of 1 foot per square second. 1 poundal equals 0.14 N

A pound is the force applied on a mass of 1 pound by the acceleration due to gravity. It is a non-technical unit of force and is also called the weight.



CONFUSION ABOUT MASS AND WEIGHT !**MASS**

- Mass is the amount of matter in the body.
- Constant regardless of the gravitational field strength
- It is a scalar quantity
- Expressed in unit - Kilograms
- Measured by a beam balance or calibrated electronic balance

WEIGHT

- Weight occurs due to the pull of the gravity.
- Varies according to the gravitational field strength
- It is a vector quantity(force)
- Expressed in Newton.
- Usually measured by a spring balance or compression balance

***If you have a mass of 60 kg , your weight on Earth is almost 600N
Your weight will differ in a different planet but your mass remains same
Weight will be 0 in space!***



WEIGHT - THE FORCE EXERTED BY THE GRAVITY

Force = mass x acceleration

Force = mass x acceleration due to gravity



- Force exerted by gravity on a body of mass 1 kg = $1\text{kg} \times 9.8\text{ms}^{-2}$
- Force exerted by gravity on a body of mass 5 kg = $5\text{kg} \times 9.8\text{ms}^{-2}$
- Force exerted by gravity on a body of mass 50 kg = $50\text{kg} \times 9.8\text{ms}^{-2}$

Acceleration due to gravity remains same on Earth. Thus, the force exerted by gravity increases with increase in mass



WANT A WEIGHT LOSS REALLY ?



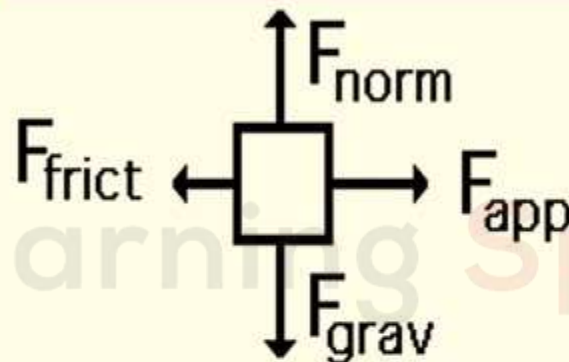
- When we diet, we tend to reduce the total amount of matter in our body.
- Thus, we tend to reduce our mass but we generally consider it as 'Weight-loss'.
- If anyone gets really interested to lose their 'weight', technically they need to move to some other planet or even can stay in space!

FREE BODY DIAGRAMS FOR DIFFERENT TYPES OF FORCES

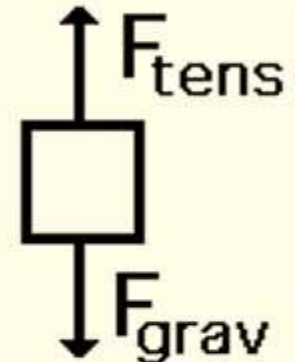
A book kept on a table



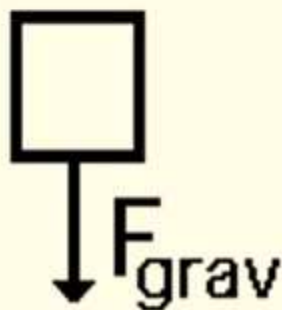
Pushing a book



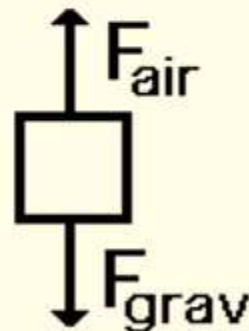
A rope pulled up



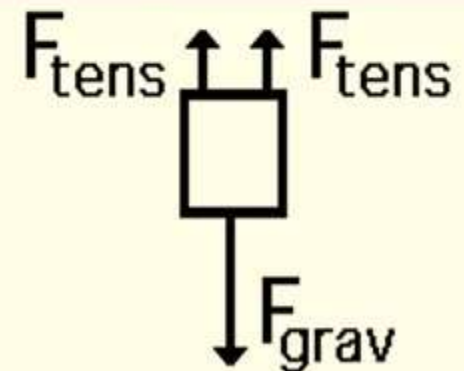
A free falling object



A flying object

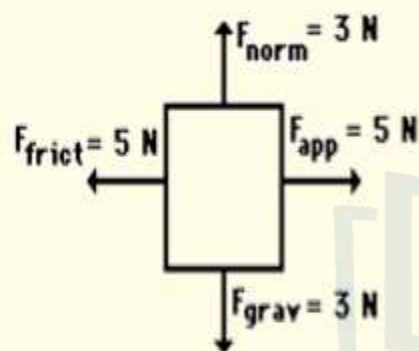


Pulling a tied rope

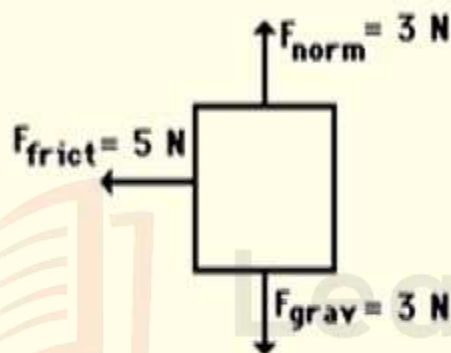


CALCULATING THE NET FORCE ACTING ON AN OBJECT

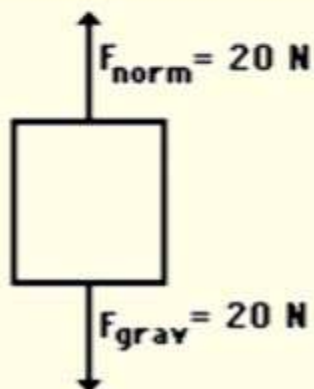
Situation A



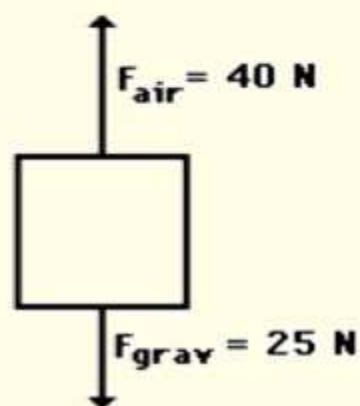
Situation B



Situation C



Situation D



- Net force acting on the object for situation A is $(3 + 5 - 3 - 5) = 0 \text{ N}$
- Net force acting on the object for situation B is $(3 + 5 - 3) = 5 \text{ N}$ as friction or drag
- Net force acting on the object for situation C is $(20 - 20) = 0 \text{ N}$
- Net force acting on the object for situation D is $(40 - 25) = 15 \text{ N}$ up

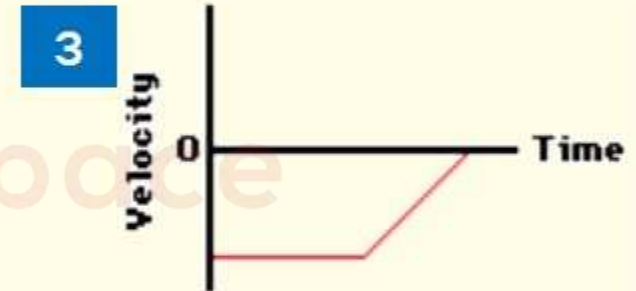




Question 1

A 5.0 kg box is dropped (weight = 50.0 N) off the roof into the swimming pool below. Upon encountering the pool, the box encounters a 50.0 N upward resistance force .

Which of the velocity time graph best describes the motion of the box?



Graph 2 is correct. The box first accelerates with a negative (downward) acceleration until it hits the water.

Upon hitting the water, the box experiences a balance of forces (50 N downwards due to gravity and 50 N upwards due to the water).

Thus, the box will finish its motion moving with a constant velocity. Graph B depicts both the initial negative acceleration and the final constant velocity.





Question 2

If the forces acting upon an object are balanced, then the object

- a) must not be moving.
- b) must be moving with a constant velocity.
- c) must not be accelerating.
- d) none of these



Balanced forces can act on moving and stationary objects and thus the body can be perpetually at rest or perpetually at motion but definitely not accelerate. Unlike it, unbalanced forces will cause movement and even acceleration to an already moving object.

Thus the correct option is C



WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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